

PRIVATE INDIAN SPACEFLIGHT

Thanks to pioneering startups and the encouragement of ISRO, private Indian spaceflight is poised to take off

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ISRO has a number of interplanetary missions and satellite launches lined up. Private spaceflight companies want in on the action too, and like ISRO, are planning their own ambitious interplanetary missions to the Moon and Venus. When it comes to satellites, ISRO cannot possibly meet the escalating demand for space technologies, and is encouraging the participation of the private sector to build up India's space capabilities.

BELLATRIX AEROSPACE

Probably India's only company working on rocket launch vehicles and propulsion systems, Bellatrix Aerospace has huge plans for the future. They have applied for a patent on their design of a microwave electrothermal thruster (MET) that uses water as the fuel. Back in April 2018, we'd covered in detail how the thruster works and the challenges they faced during the entire course of the project until the first successful prototype. (<https://digit.in/BellatrixAero>). This propulsion system is highly scalable and directed towards satellites only because of the low thrust generated. The team is still working out on various components of the system, eventually moving on to testing the MET in its full form.

Bellatrix has received funding for their MET project while they are still awaiting to raise funds for their secondary projects involving launch vehicles. Currently, they have two designs, Chetak and Garuda, that are capable



The TeamIndus moon rover

of carrying a payload of 500kg and 1,010kg respectively. The MET isn't the only propulsion system they are working on. Working on Hall Thrusters, they are trying to achieve a longer lifespan than the traditional designs. They're also trying out their hands in developing a cleaner alternative to Monopropellant Hydrazine Thrusters used in chemical propulsion systems. It's a toxic chemical, so the company is working on the alternative and they are calling it Green Monopropellant Thruster (HAN blend).

ISRO

Antrix, the commercial arm of ISRO is actually a private spaceflight company.

Antrix provides among the cheapest and most reliable launch services in the world, primarily through its workhorse rocket – the PSLV. It is not a very big or very powerful rocket, but this actually worked to its advantage because of the miniaturisation of satellites. Infact, ISRO is working on developing a smaller rocket specifically to launch nanosatellites and microsatellites. Additionally, ISRO works closely with private industries in the country for sub-systems used in its launch vehicles and satellites. ISRO intends to transfer its technologies to private companies, and privatise at least portions of the rocket and satellite production.